

UNIVERSITY PARTNER



4MM013 - Computational Mathematics

Mathematics Assignment-1

Full Marks: 10

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1. State the definition of a function and a composite function.

[2Marks]

Let f and g be functions defined as follows:

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{x-3}{x+1}, f(2) = ? \text{ and}$$

$$g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \frac{1}{x}, x \neq 0$$

Calculate $(f \circ g)(x)$ and $(g \circ f)(x)$.

A rule which provides every input value a distinct output value is a function which is to say one input value is assigned with an output value. It takes an input, processes it and gives output as result. If there is a function ' f ' which adds number 3 to the input then if input is 2 the output becomes 5. Function is written as $f(x)$, where x is the input value. So, in this case $x=2$ and output is 5.

The combination of two or more functions can be regarded as composite function. If there were two functions $f(x)$ and $g(x)$. Then the composite function would be denoted as $(f \circ g)(x)$, which is pronounced as f composed with g of x . When $f(x)=3+x$ and $g(x)=2x$. Then $(f \circ g)(x)$ would be $f(g(x))=3+2x$.

Handwritten solution for the composite function problem:

1. Here, $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{x-3}{x+1}, f(2) = ?$

Now, inputting the value of x as 2,

$$f(2) = \frac{2-3}{2+1}$$
$$= \frac{-1}{3} \text{ Ans}$$

$g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \frac{1}{x}, x \neq 0.$

Now, $f \circ g(x)$

Here, $f(x) = \frac{x-3}{x+1}$

$$g(x) = \frac{1}{x}$$
$$f(g(x)) = f\left(\frac{1}{x}\right) = \frac{\frac{1}{x}-3}{\frac{1}{x}+1}$$
$$= \frac{1-3x}{1+x} \text{ Ans}$$

$$\begin{aligned}
 g \circ f(n) &= g\left(\frac{n-3}{n+1}\right) \\
 &= \frac{1}{\frac{(n-3)}{n+1}} = \frac{n+1}{n-3} \text{ Ans}
 \end{aligned}$$

2. Solve the following using the inverse matrix method:

[2 Marks]

$$6x - y = 0$$

$$2x - 4y = 1$$

2.

Here,

$$6x - y = 0$$

$$2x - 4y = 1$$

$$A = \begin{bmatrix} 6 & -1 \\ 2 & -4 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1} \cdot B$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj}(A)$$

$$\begin{aligned} \text{Now, } |A| &= 6 \times (-4) - (-1) \times 2 \\ &= -24 + 2 \\ &= -22 \neq 0 \end{aligned}$$

$$\text{Adj}(A) =$$

$$A_{11} = -1$$

$$A_{12} = -2$$

$$A_{21} = 1$$

$$A_{22} = 6$$

$$\text{Co-factors of matrix } A = \begin{bmatrix} -1 & -2 \\ 1 & 6 \end{bmatrix}$$

$$\text{Now, Transpose} = [A]^T = \begin{bmatrix} -1 & 1 \\ -2 & 6 \end{bmatrix} \text{ which is } \text{Adj}(A)$$

$$\begin{aligned} \text{Then, } A^{-1} &= \frac{1}{|A|} \text{Adj}(A) \\ &= \frac{-1}{22} \begin{bmatrix} -1 & 1 \\ -2 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 2/11 & -1/22 \\ 1/11 & -3/11 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} X &= A^{-1} \cdot B \\ &= \begin{bmatrix} \frac{2}{11} & \frac{-1}{22} \\ \frac{1}{11} & \frac{-3}{11} \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} &= \begin{bmatrix} 2/11 \times 0 + (-1/22) \times 1 \\ 1/11 \times 0 + (-3/11) \times 1 \end{bmatrix} \\ &= \begin{bmatrix} -1/22 \\ -3/11 \end{bmatrix} \end{aligned}$$

Therefore, $X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1/22 \\ -3/11 \end{bmatrix}$

Thus, $x = -1/22$ & $y = -3/11$

3. Calculate the inverse of the following functions:

[2 Marks]

a. $f(x) = \frac{6+x}{7}$

3.
a. $f(x) = \frac{6+x}{7}$

let $f(x) = y$,

$$y = \frac{6+x}{7}$$

$$7y = 6+x$$

$$x = 7y - 6$$

Replacing y with $f^{-1}(x) = 7x - 6$

Therefore, $f^{-1}(x) = 7x - 6$

b. $f(x) = \frac{3}{2-x}$

$$b. f(x) = \frac{3}{2-x}$$

Replacing $f(x)$ with y ,

$$y = \frac{3}{2-x}$$

$$2y - yx = 3$$

$$x = \frac{2y-3}{y}$$

Replacing y with $f^{-1}(x)$

$$\therefore f^{-1}(x) = \frac{2x-3}{x}$$

4. Sketch the graph of the following functions:

[2 Marks]

$$f(x) = 2(x-1)^2 + 3, -2 < x < 2 \quad \text{In the interval of } x=0.5$$

4.

Setting derivative of $f(x) = 0$ and solving x ,

$$f'(x) = 4(x-1)$$

$$4(x-1) = 0$$

$$x = 1$$

Putting the value of x in above equation,

$$f(1) = 2(1-1)^2 + 3$$

$$= 3$$

Hence, vertex of parabola is $(1, 3)$.

Here, coefficient of squared term being positive, the parabola opens upwards.

Now, y -intercept when $x = 0$,

$$f(0) = 2(0-1)^2 + 3$$

$$= 5$$

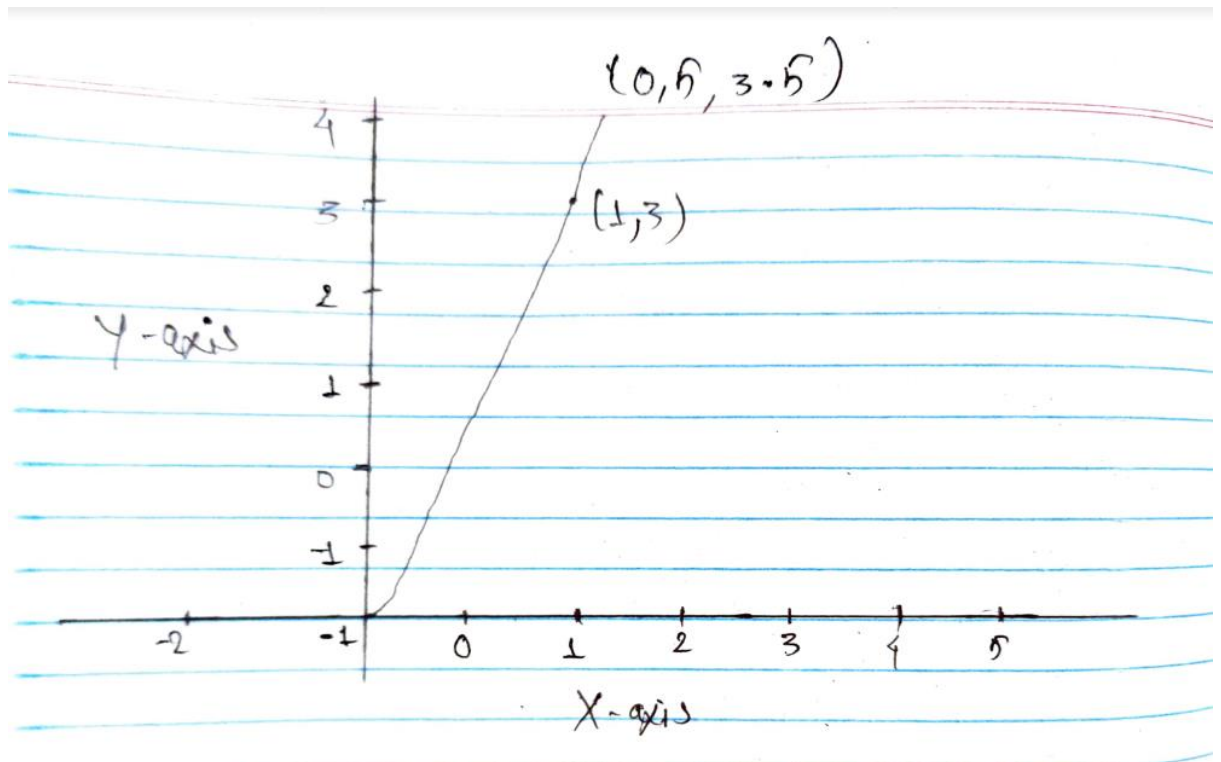
Thus, y -intercept $= (0, 5)$

Interval $= 0.5$,

Putting $x = 0.5$

$$f(0.5) = 2(0.5-1)^2 + 3$$

$$= 3.5$$



5.

a. Define gradient of a function. State the gradient and intercept of :

$$2y + 8 = 6x$$

5.

a.

$2y + 8 = 6x$, Arranging the function in the form of $y = mx + c$

$$2y = 6x - 8$$

$$2y = 2(3x - 4)$$

$$y = 3x + (-4)$$

Comparing the above function to $y = mx + c$.
gradient, $m = 3$
& intercept, $c = -4$

[2 Marks]

b. Solve the following equations:

$$x^2 + \frac{17}{6}x + 2 = 0$$

b. $x^2 + \frac{17}{6}x + 2 = 0$

Here, solving the above quadratic equation which is in the form of $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here, submitting the values of a , b & c in the above equation,

$$= \frac{-17/6 \pm \sqrt{(17/6)^2 - 4 \times 1 \times 2}}{2 \times 1}$$

$$= \frac{-17/6 \pm \sqrt{(289 - 288)/36}}{2}$$

$$= \frac{-17/6 \pm 1/6}{2}$$

Now, solving for x ,

$$x = \frac{(-17/6 + 1/6)}{2} = \frac{(-16/6)}{2} = \frac{-4}{3}$$

$$x = \frac{(-17/6 - 1/6)}{2} = \frac{(-18/6)}{2} = \frac{-3}{2}$$

The End